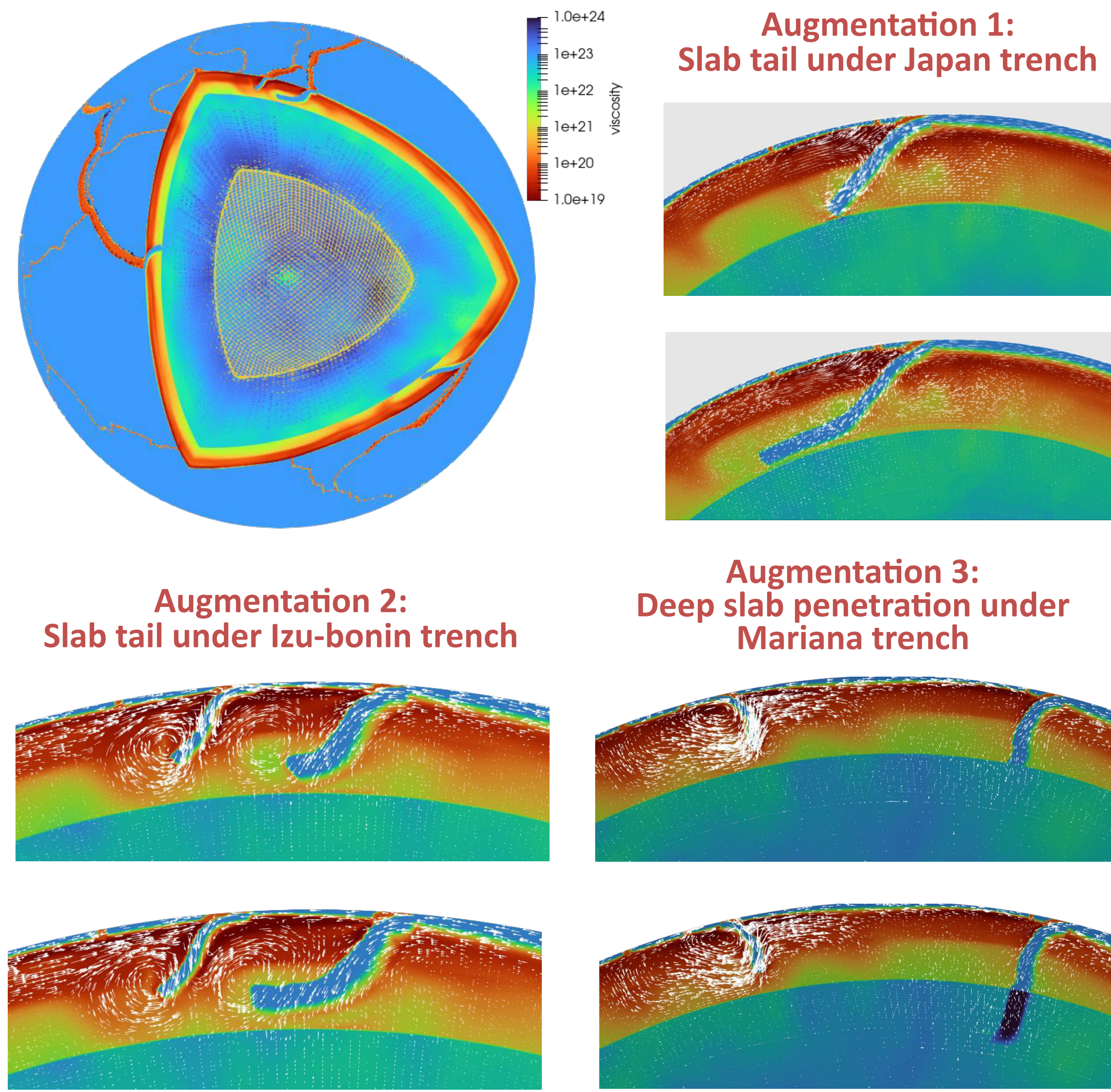


1. Motivation

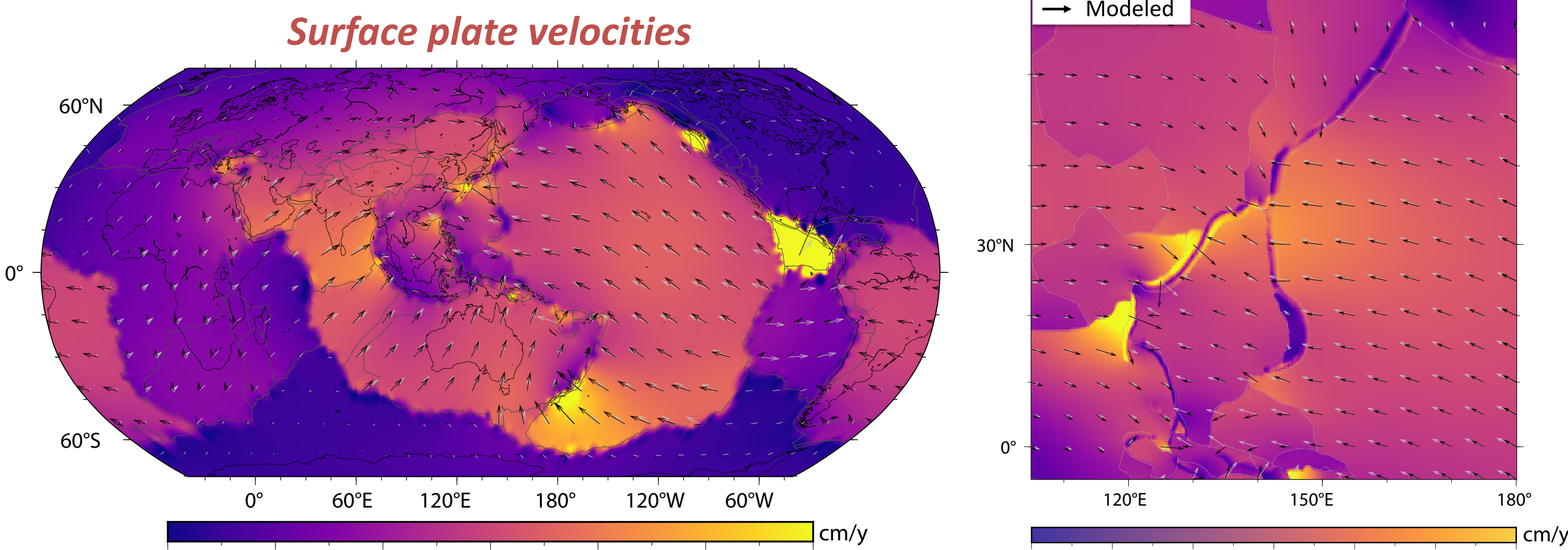
- The strength and direction of mantle flow exert a significant control on important subduction zone processes, such as near-surface tectonic deformation, and conversely are themselves governed by subduction processes.
- Our understanding of such plate-mantle interactions, especially in complex tectonic regions like the western Pacific, is limited.
- E.g., the Philippine Sea Plate (PSP) is surrounded by slabs and is shrinking. How does mantle flow organize beneath the PSP?
- We investigate this using global-scale mantle convection models with slabs, weak plate boundaries, and a high-resolution western Pacific.

2. Numerical model setups

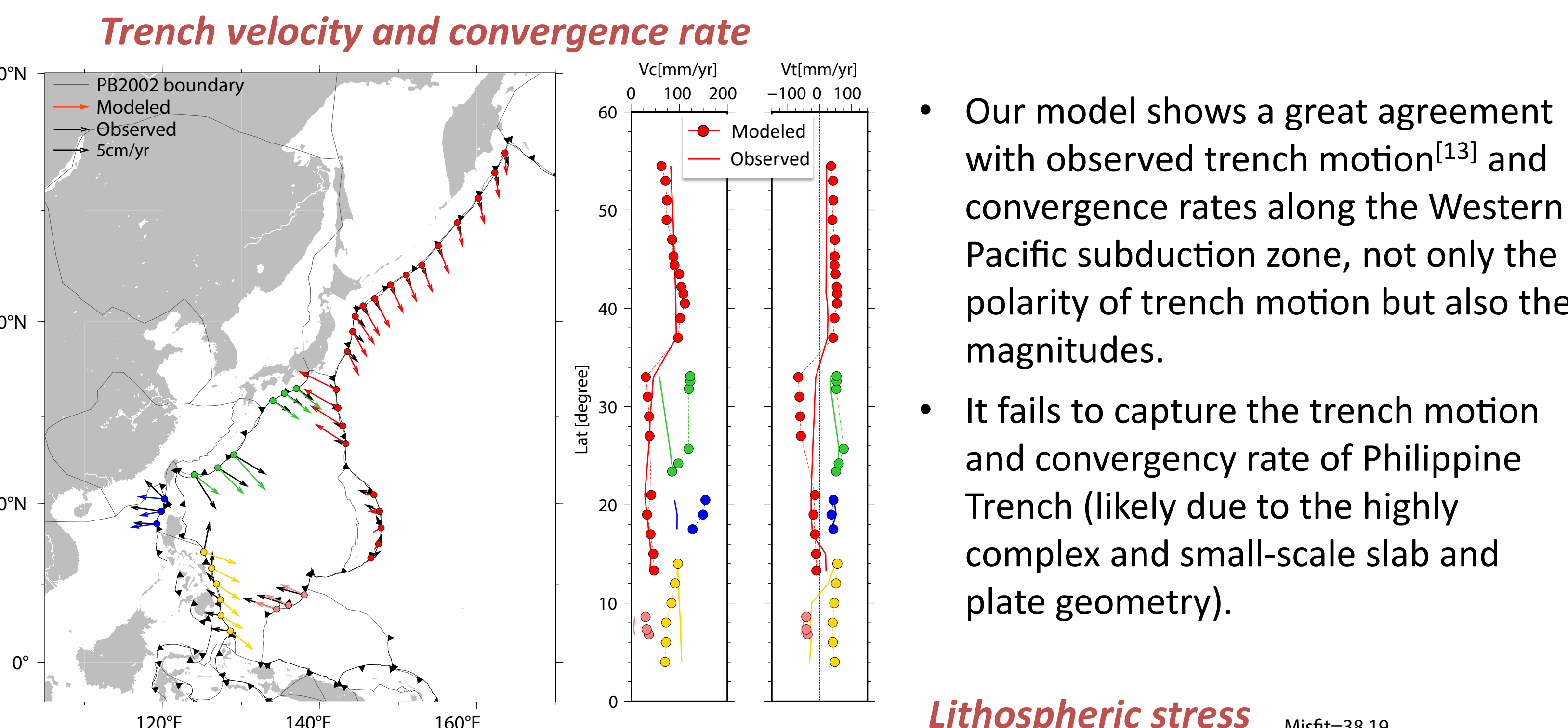
- Overall approach:** 3-D instantaneous global models^[1] with locally high resolution in the Western Pacific region constructed using the ASPECT finite element code^[2-5].
- Thermal structure**
 - Background mantle: from seismic tomography^[6-7].
 - Lithosphere: from half-space cooling and seismic observations^[8].
 - Slabs: realistic shaped slabs from the Slab2 and tomography^[7,9-11].
- Mantle Rheology**
 - Diffusion creep (whole mantle) and dislocation creep (upper mantle).
 - Weak plate boundaries (30km) go along with slab top.
- Model ensemble (4 mantle structure tests)**
 - Background mantle structure (tomography model outside of slabs). (1) SAVANI^[6]; (2) REVEAL^[7].
 - Vary slab structure. (1) SLAB2^[9]; (2) SLAB2 augmented with tomography models^[7,10,11].
 - Reference model in Box3 and Box 5: SAVANI with augmented SLAB2.



3. Model validation

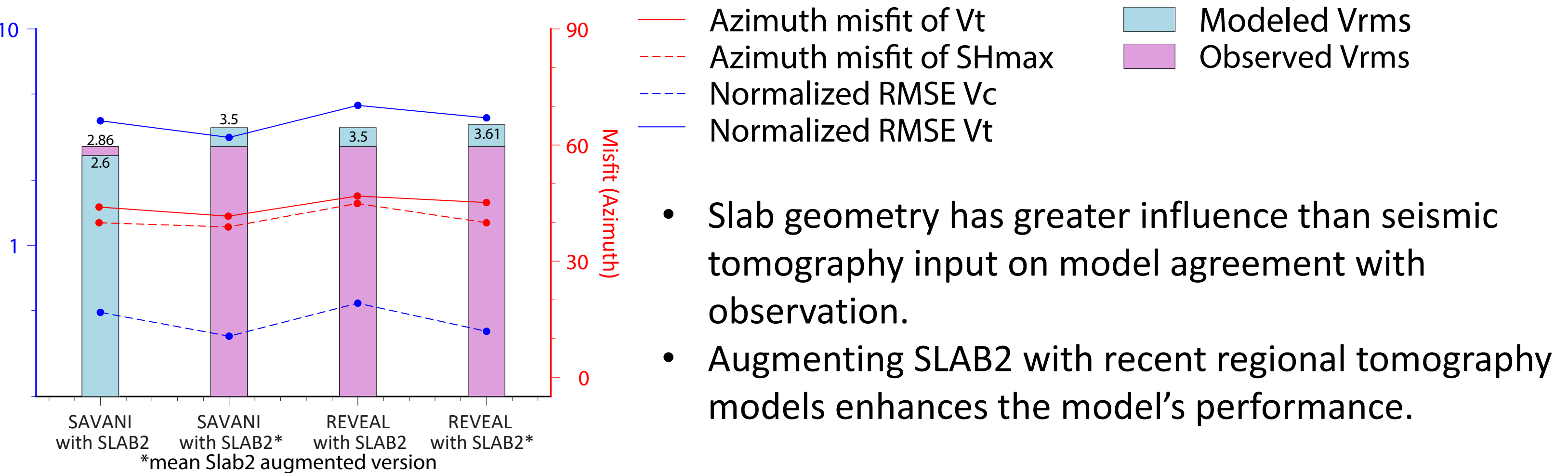


- Model matches plate motions^[12] to first order, including the Pacific and Australian Plates.
- In the western Pacific, we capture the dominant plate motion directions and magnitudes.

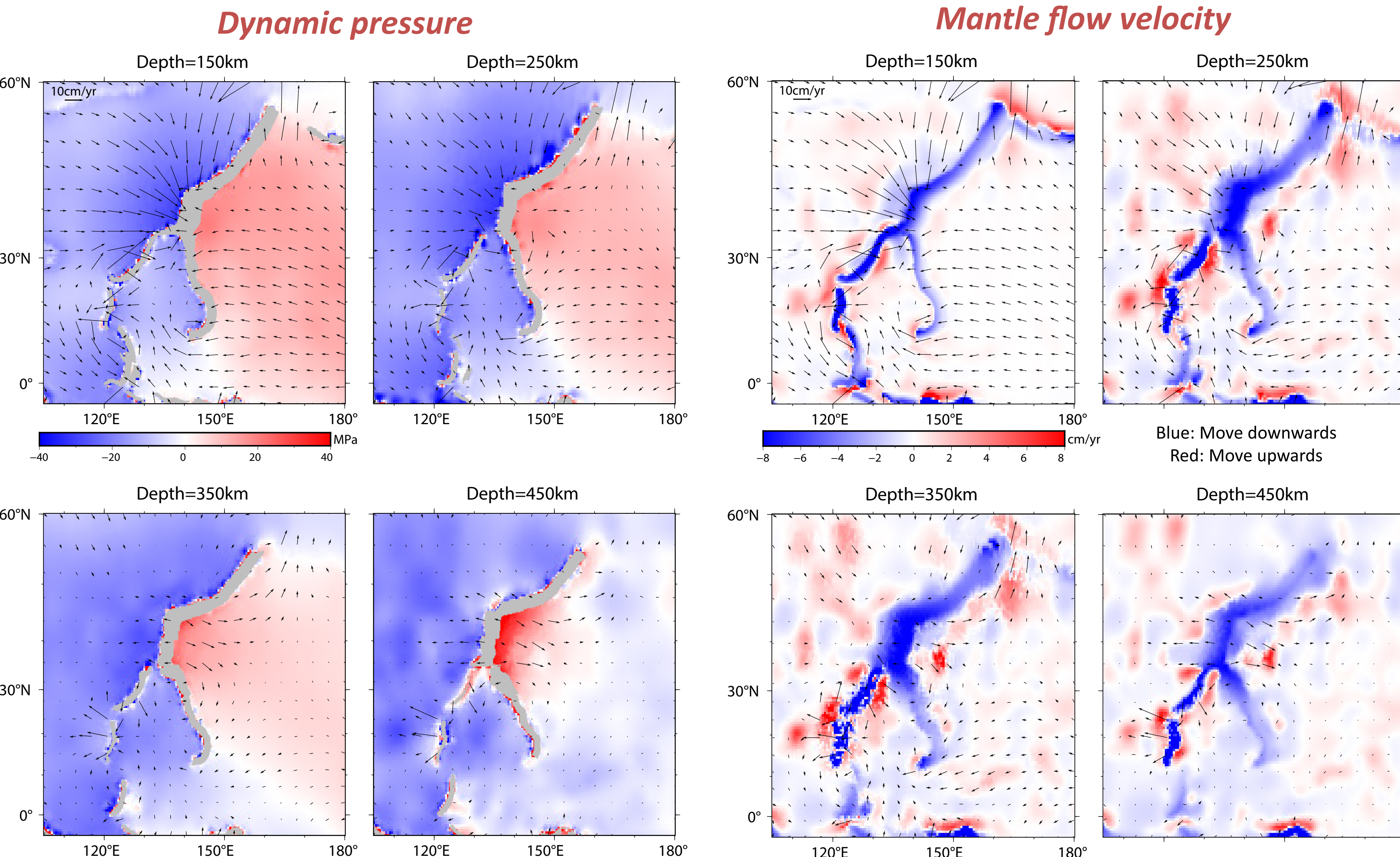


- Our model shows a great agreement with observed trench motion^[13] and convergence rates along the Western Pacific subduction zone, not only the polarity of trench motion but also the magnitudes.
- It fails to capture the trench motion and convergence rate of Philippine Trench (likely due to the highly complex and small-scale slab and plate geometry).
- Our models have maximum principal stress directions that are similar to the World Stress Map^[14-15] near to subduction zones.
- However, we fail to accurately reproduce the stress distribution in continental regions or regions of especially complex plate boundary geometries (e.g., Yangtze plate).

4. Model comparison



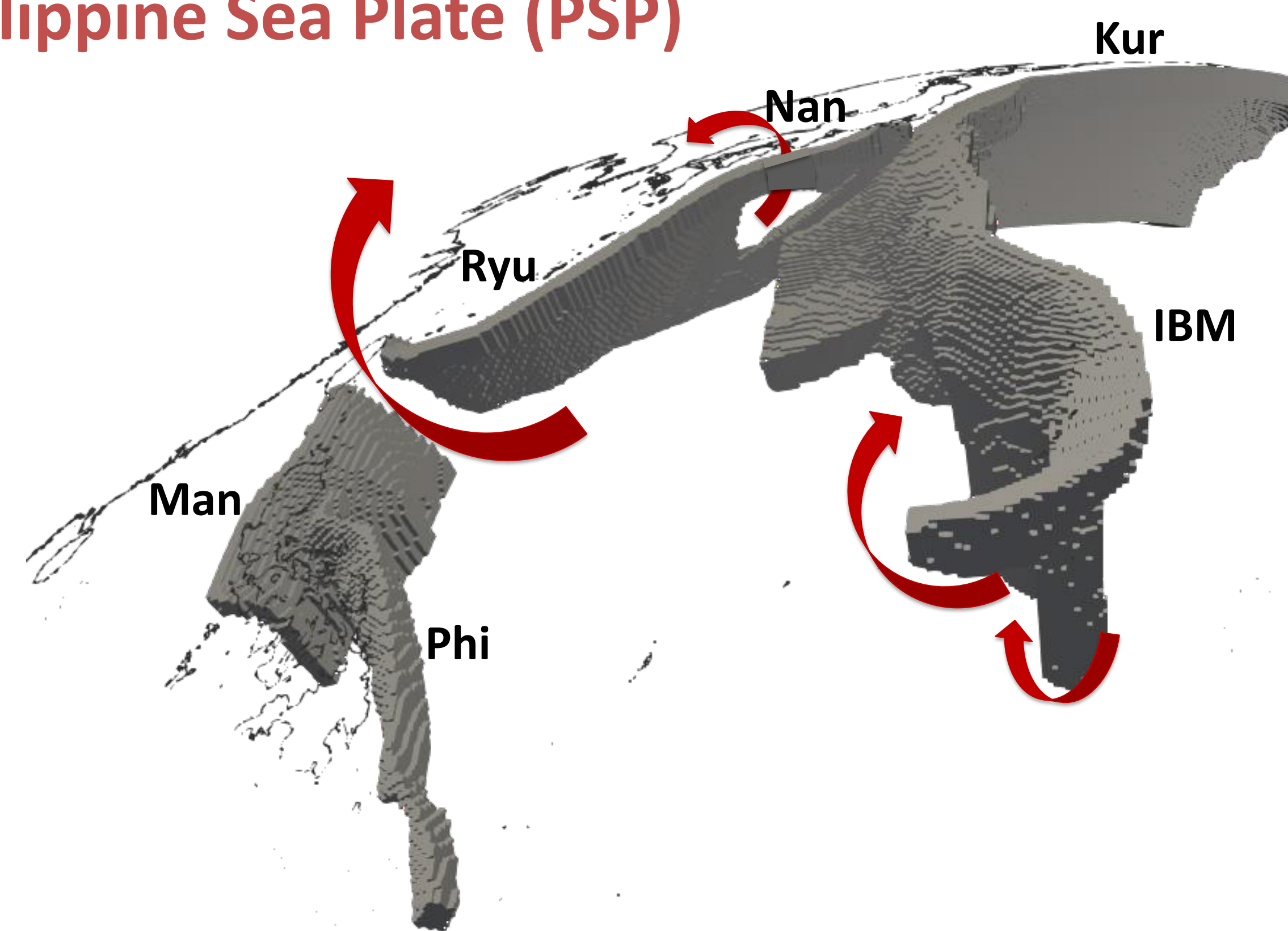
5. Mantle pressure and flow velocity



- In single Subduction zones (e.g., Japan/Kuril), pressure pattern is consistent with idealized subduction models^[16] (i.e., $P_{wedge} - P_{sub-slab} < 0$).
- In double Subduction zone (IBM/Ryukyu/Manila), the pressure drop is also negative. This deviates from idealized models of double subduction but aligns with the absence of overturned slabs.
- Toroidal flow develops around slab edges, driven by pressure gradients. Toroidal flow is strongest around the Ryukyu slab.
- Poloidal flow develops in front of the slabs beneath the Izu-Bonin trench and Ryukyu/Nankai trenches. (Also see figures in Box 2).

6. Flow into/out of Philippine Sea Plate (PSP)

- Coupled with trench motion, toroidal flow show in-flow mode in the southern edge of PSP, out-flow pattern around the northern edge.
- Out-flow around Ryukyu and Nankai trenches facilitates the escape of mantle material from the PSP, contributing to its shrinking.



Kur: Kuril Trench; IBM: Izu-bonin-Mariana Trench; Ryu: Ryukyu Trench; Nan: Nankai Trench; Man: Manila Trench; Phi: Philippine Trench

Acknowledgments

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